

Prepared by Prachi Sarode



Medical Diagnostic AI

Classification Models, Evaluation Metrics & Handling Imbalanced
Data

Project Overview & Objectives

Transitioning from Regression to Clinical Binary Classification

Industry Context & Scope

- **Binary Problem:** Classifying tumors as Malignant (0) vs. Benign (1) using 30 clinical features.
- **Accuracy Illusion:** In imbalanced datasets, simple accuracy gives a false sense of model security.
- **High-Stakes Decisioning:** Misclassifying a malignant case leads to severe real-world consequences.
- **Deployment Focus:** Building robust pipelines tailored for clinical evaluation.



Source: GAO. | GAO-22-104629

Data Pipeline & Preprocessing

Stratified Splitting

Splitting dataset with an 80/20 train-test ratio while enforcing target stratification. This guarantees proportional representation of both malignant and benign target classes across splits.

Feature Standardization

Applying StandardScaler to fit training parameters and transform test features. Ensures zero mean and unit variance across all 30 dimensional feature vectors.

Confusion Matrix Analysis

Actual Class \ Predicted	Predicted Malignant (0)	Predicted Benign (1)	Clinical Impact Analysis
Actual Malignant (0)	41 (True Positives)	1 (False Negative)	Critical FN: Missed 1 cancer patient (High Risk)
Actual Benign (1)	1 (False Positive)	71 (True Negatives)	Low FP: 1 patient needs extra follow-up testing

Model Discrimination Ability

0.997

ROC-AUC Score

Near-Perfect Separability

The Baseline Logistic Regression model achieves an outstanding Area Under the ROC Curve (AUC = 0.9974). This proves the model's exceptional capability to differentiate between benign and malignant classes across all classification probability thresholds.

| Key Evaluation Metrics



Precision (98%)

Measures how trustworthy positive predictions are. Minimizes false alarms in screening tests.



Recall (98%)

Top priority metric for diagnosis. Minimizes dangerous False Negatives by identifying actual malignant cases.



F1-Score (0.98)

Harmonic mean balancing precision and recall, essential when working with imbalanced data distributions.

Model Performance Comparison



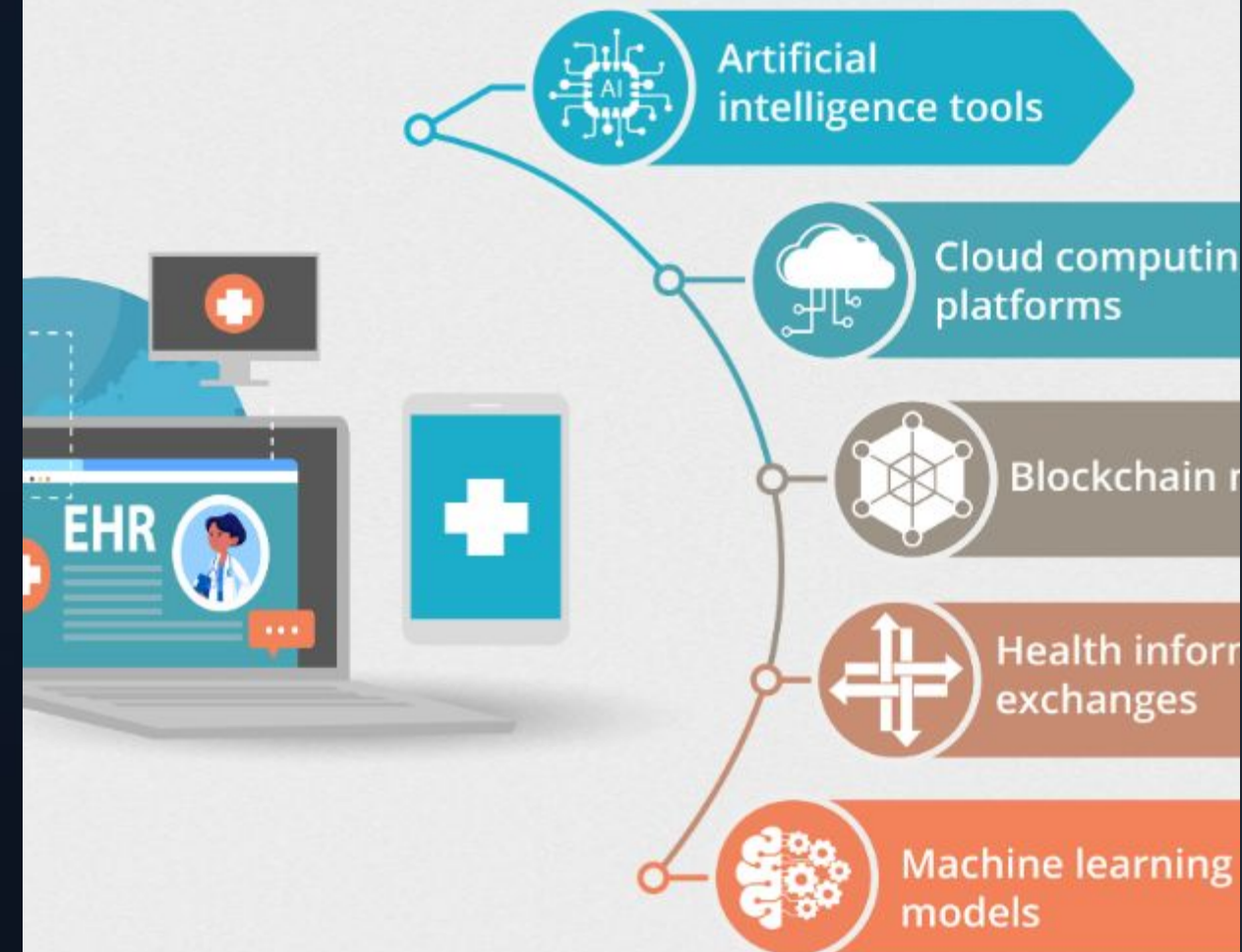
Logistic Regression outperforms Decision Trees in stability and generalization, achieving higher overall accuracy (98% vs 94%) and robust probability calibration.

Handling Class Imbalance

Class-Weighted Training

In medical datasets, malignant cases are often minority samples. By setting `class_weight='balanced'` in Logistic Regression, the model automatically adjusts loss weights inversely proportional to class frequencies. This penalizes misclassifying rare malignant cases heavily, optimizing model Recall without reducing accuracy.

5 TYPES OF HEALTHCARE DATA ANALYTICS TECHNOLOGIES



Execution Roadmap

